Numericals

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Resistivity and Resistance

1. A rectangular bar of iron is $2.0\ cm$ by $2.0\ cm$ in cross section and $40\ cm$ long. Calculate its **resistance**, If the resistivity of iron is $11 \times 10^{-8}\Omega m$.

Given:

- $L = 40cm = 0.4m$
- $A = 2.0cm \times 2.0cm = 4.0cm^2 = 4 \times 10^{-4}m^2$
- $\rho = 11 \times 10^{-8}\Omega m$

To Find:

$R = ?$

Formula:

$$R = \rho\frac{L}{A}$$

Solution:

$$R = \frac{11 \times 10^{-8} \times 0.4}{4 \times 10^{-4}}$$

$$R = 1.1 \times 10^{-4}\Omega$$

Electric Current

2. The current in the electron beam of a typical video display terminal is $200\mu A$. **How many electrons strike the screen each minute**?

Given:

- $i = 200 \times 10^{-6}A$
- $t = 60s$
- $e = 1.6 \times 10^{-19}C$

To Find:

$n = ?$

Formula:

- $n = \frac{q}{e}$
- $q = it$

Solution:

$q = 200 \times 10^{-6} \times 60$

$q = 1.2 \times 10^{-2}C$

$$n = \frac{1.2 \times 10^{-2}}{1.6 \times 10^{-19}}$$

$n = 7.5 \times 10^{16} electrons$

Ohm's Law and Power

3. A heating wire has a resistance 72Ω and it is also connected to a battery of $120V$. Under what condition will it **dissipate more heat**?

(a) Its length is connected across a battery.

(b) The wire is cut in half and two halves are connected in parallel across battery.

Given:

- $V = 120V$
- $R = 72\Omega$

To Find:

$\mathbf{P} = ?$

Formula:

- $P = \frac{V^2}{R}$
- *Since wire is cut in 2 halves,*

$P = \frac{V^2}{\frac{R}{2}}$

$P = \frac{2V^2}{R}$

Solution:

$(\mathbf{a})\ P = \frac{(120)^2}{72}$

$P = 200W$

$(\mathbf{b})\ P = \frac{2(120)^2}{72}$

$P = 400W$

Since, wire is split in 2 halves

Total power=400+400=800W

The wire dissipates more heat in **condition (b)** (parallel connection), where the total power is $\mathbf{800W}$, which is **4 times** the power of the single wire

Forces in Magnetic/Electric Fields

4. An electric field of $1.5\ kV/m$ and a magnetic field of $0.44T$ act on a moving electron to produce no force. Find **minimum speed of electron**.

Given:

- $E = 1.5 \times 10^3\ kV/m$
- $B = 0.44T$

To Find:

$v = ?$

Formula:

$v = \frac{E}{B}$

Solution:

$v = \frac{1.5 \times 10^3}{0.44}$

$v = 3400ms^{-1}$

Magnetic Field of a Solenoid

5. A solenoid is $1.33m$ long and diameter $2.6cm$ carries as current of $17.8A$. The magnetic field inside the solenoid is $22.4mT$. Find the **total number of turns of solenoid**.Magnetic Field of a Solenoid

Given:

- $l = 1.33m$
- $i = 17.8A$
- $B = 22.4mT = 22.4 \times 10^{-3}T$
- $\mu_o = 4\pi \times 10^{-7}\ Tm/A$

To Find:

$N = ?$

Formula:

$N = \frac{Bl}{\mu_o I}$

Solution:

$N = \frac{22.4 \times 10^{-3} \times 1.33}{4\pi \times 10^{-7} \times 17.8}$

$N = 1332$

Magnetic Field due to a Long Straight Wire

6. Find value of **magnetic field** at a distance $2cm$ from a long straight conductor in which current $2mA$ flows.

Given:

- $R = 2 \times 10^{-2}m$
- $i = 2 \times 10^{-3}A$
- $\mu_o = 4\pi \times 10^{-7}\ Tm/A$

To Find:

$B = ?$

Formula:

$$B = \frac{\mu_o i}{2\pi R}$$

Solution:

$$B = \frac{4\pi \times 10^{-7} \times 2 \times 10^{-3}}{2 \times 3.14 \times 2 \times 10^{-2}}$$

$$B = 2 \times 10^{-8}T$$

Motional EMF

7. An automobile having radio antenna $110cm$ long travels at $90\ km/h$ in a region where earth magnetic field is $55\mu T$. Find the **maximum value of induced emf**.

Given:

- $L = 110cm = 1.10m$
- $B = 55\mu T = 55 \times 10^{-6}T$
- $v = 90\ km/h = \frac{90 \times 1000}{60 \times 60} = 25ms^{-1}$

To Find:

$\varepsilon = ?$

Formula:

$\varepsilon = vBL$

Solution:

$\varepsilon = 25 \times 55 \times 10^{-6} \times 1.10$

$\varepsilon = 1.5 \times 10^{-3}V$

Magnetic Force on a Charge

8. A proton travelling at 23° w.r.t. magnetic field of strength $2.63mT$ experiences a magnetic force of $6.48 \times 10^{-17}N$. Find the **speed of proton**.

Given:

- $\theta = 23^{\circ}$
- $B = 2.63mT = 2.63 \times 10^{-3}T$
- $F = 6.48 \times 10^{-17}N$
- $q = 1.6 \times 10^{-19}C$

To Find:

$v = ?$

Formula:

$\vec{F} = q(\vec{v} \times \vec{B})$

$F = qvbsin\theta$

$$v = \frac{F}{qBsin\theta}$$

Solution:

$$v = \frac{6.48 \times 10^{-17}}{1.6 \times 10^{-19} \times 2.63 \times 10^{-3} \times \sin 23^{\circ}}$$

$v = 3.94 \times 10^{5}ms^{-1}$

Inductance

9. The current amplitude in a pure conductor in a radio receiver is to be $250\mu A$ when the voltage amplitude is $3.6V$ at a frequency of $1.6MHz$ (at the upper end of AM broadcast band). What **inductance reactance** is needed? What is **inductance**?

Given:

- $I = 250\mu A = 250 \times 10^{-6} A$
- $V = 3.6V$
- $f = 1.6MHZ = 1.6 \times 10^{6} Hz$

To Find:

- $X_L = ?$
- $L = ?$

Formula:

- $X_L = \dfrac{V}{I}$
- $L = \dfrac{X_L}{\omega}$

  $L = \dfrac{X_L}{2\pi f}$

Solution:

$X_L = \dfrac{3.6}{250 \times 10^{-6}}$

$X_L = 1.44 \times 10^{4} \Omega$

$L = \dfrac{1.44 \times 10^{4}}{2 \times 3.14 \times 1.6 \times 10^{6}}$

$L = 1.43 \times 10^{-3} H$

Wave Properties

10. A sinusoidal electromagnetic wave of frequency $40MHz$ travels in free space in the x-direction. Determine the **wavelength** and **period of the wave**.

Given:

- $c = 3 \times 10^{8} T$
- $f = 40MHz = 40 \times 10^{6} Hz$

To Find:

- $\lambda = ?$
- $T = ?$

Formula:

- $\lambda = \dfrac{c}{f}$
- $T = \dfrac{1}{f}$

Solution:

$\lambda = \dfrac{3 \times 10^{8}}{40 \times 10^{6}}$

$\lambda = 7.5m$

$T = \dfrac{1}{40 \times 10^{6}}$

$T = 2.5 \times 10^{-8} s$

Speed of Light

11. The distance between Earth and Sun is $1.5 \times 10^{11} m$. **How long does it take for light to cover this distance**?

Given:

- $S = 1.5 \times 10^{11} m$
- $c = 3 \times 10^{8} ms^{-1}$

To Find:

$t = ?$

Formula:

$t = \dfrac{S}{v}$

$t = \dfrac{S}{c}$

Solution:

$$t = \frac{1.5 \times 10^{11}}{3 \times 10^{8}}$$
$$t = 500s$$